

Decoding as Alignment: Assessing Decoding Strategies as a Practical Alternative to Full RLHF in Low-Resource Korean LLM Settings

Dongyeun Kum

MODULABS

March 2026

Abstract

Reinforcement Learning from Human Feedback (RLHF) is the standard for aligning LLMs, but its high cost is a barrier in low-resource settings. This paper investigates inference-time decoding optimization as a practical alternative for small-scale Korean LLMs. Using KoGPT-2, we show that a Supervised Fine-Tuned (SFT) model with tuned decoding matches the reward scores of a PPO-trained model while achieving significantly higher response diversity (Distinct-2: 0.864 vs. 0.668). We additionally identify a strong length bias ($r = 0.719$) in the reward model, suggesting that its reliability is limited in low-resource settings and further motivating inference-time alternatives to full RLHF pipelines. Our findings indicate that decoding strategy optimization offers a cost-effective trade-off between alignment quality and computational cost under hardware and data constraints.

1 Introduction

Aligning Large Language Models (LLMs) with human intent is a key step in making them helpful and safe. The standard approach is Reinforcement Learning from Human Feedback (RLHF), which trains a reward model and optimizes a language model using Proximal Policy Optimization (PPO). [1, 7] Although effective, RLHF is computationally expensive and requires complex data pipelines, making it difficult for individual researchers or small teams to adopt.

This raises a practical question: is full PPO training always necessary for alignment in low-resource settings? For small-scale Korean models such as KoGPT-2 [3], the overhead of RLHF may outweigh its benefits.

In this paper, we explore the inference-time decoding control as a lighter alternative. We investigate whether a Supervised Fine-Tuned (SFT) model, when paired with carefully chosen decoding strategies such as beam search and repetition penalties [4], can offer a reasonable trade-off between alignment quality and computational cost without additional training. This provides a more accessible path for aligning small models under hardware and data constraints.

2 Method

2.1 Baselines: Model and Dataset

We adopted KoGPT-2 (125M) as our base model and the KoChatGPT dataset [2] for RLHF training. The dataset provides 12,000 pairs of SFT, 10,220 RM ranking triplets, and 12,000 PPO prompts, all synthetically generated using ChatGPT and GPT-3 instead of human annotation. After cleaning (Korean ratio filtering, length filtering, encoding validation), usable samples were reduced to 11,436, 9,121, and 11,970, respectively. The high RM removal rate (10.8%) reflects the noisy nature of automatically ranked triplets in this low-resource setting.

2.2 Experimental Setup: Full RLHF Training

We independently executed the full training pipeline (SFT $\rightarrow$ RM $\rightarrow$ PPO) on a single NVIDIA Tesla T4 GPU (14.6 GB VRAM). The SFT model was trained for 3 epochs with a learning rate of $2e-5$, weight decay of 0.01, and an effective batch size of 8 (per-device batch size 2 with gradient accumulation steps of 4). Mixed precision training (fp16) was enabled to reduce memory usage. The Reward Model was trained for 1 epoch using pairwise ranking loss ($lr=5e-5$, batch size 4) on chosen/rejected pairs derived from the RM triplets. PPO training was conducted for 10 episodes with a train batch size of 8, max timesteps of 3 and an inner update epoch of 1. All three checkpoints were saved and used as-is for the decoding experiments, with no further fine-tuning.

2.3 Decoding Strategies

We evaluated six conditions: three model checkpoints (Base, SFT, PPO) each paired with two decoding strategies.

- Default: Greedy decoding (`do_sample=False`, `num_beams=1`, `repetition_penalty=1.0`), serving as the unoptimized baseline.
- Tuned: Beam search (`num_beams=4`) combined with nucleus sampling (`top_p=0.92`, `top_k=50`), temperature scaling (`temperature=0.7`), and repetition control (`repetition_penalty=1.3`, `no_repeat_ngram_size=4`)

All decoding parameters were fixed uniformly across models to ensure a fair comparison. We investigated whether SFT with Tuned decoding offers a reasonable trade-off in alignment quality relative to PPO, without the overhead of reinforcement learning training.

3 Results

3.1 Overall Performance Comparison

Figure 1 summarizes results across all six conditions. PPO+Default achieves the highest RM Reward (-1.408), with SFT+Tuned scoring -1.525 — a gap of only 0.117. This difference should be interpreted cautiously given the reward model’s limited discriminability (Section 3.4).

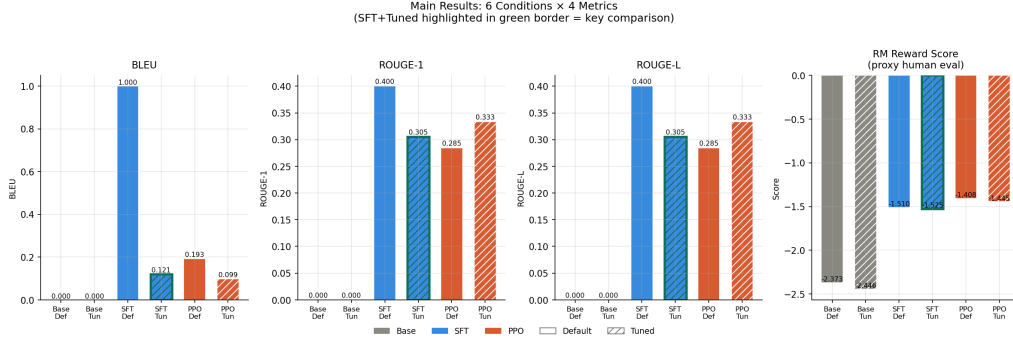


Figure 1: Average RM Reward and performance comparison across all conditions. BLEU and ROUGE are measured relative to the SFT+Default anchor; RM Reward Score serves as a proxy for human preference. SFT+Tuned (green border) is the key comparison condition.

On ROUGE-1 and ROUGE-L [9], SFT+Tuned (0.305) outperforms PPO+Default (0.285), suggesting that Tuned decoding can partially compensate for the absence of reinforcement learning in terms of response coherence. Figure 2 further confirms that both PPO conditions diverge substantially from the SFT response style (RM $\uparrow$, BLEU $\downarrow$), while SFT+Tuned preserves the instruction-following structure of the SFT baseline.

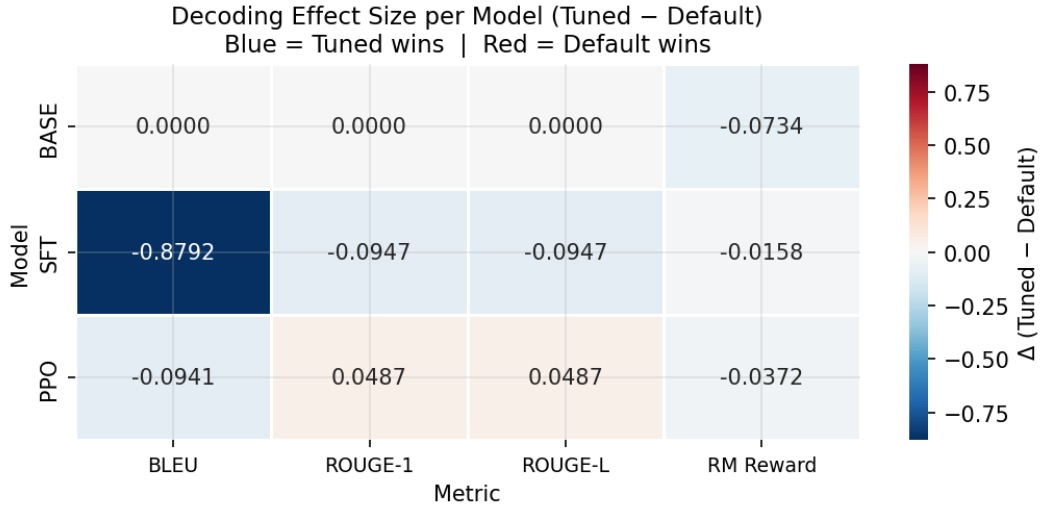


Figure 2: Decoding effect size per model (Tuned – Default). Blue = Tuned wins; red = Default wins. The large negative BLEU for SFT reflects divergence from the SFT+Default anchor, not a quality drop

3.2 Response Diversity and Lexical Richness

SFT+Tuned achieves a Distinct-2 score [5] of 0.864, compared to 0.668 for PPO+Default - a 29.3% relative improvement (Figure 3). Self-BLEU [6] further confirms lower redundancy for SFT+Tuned (0.088 vs. 0.011 for PPO+Default, where near-zero values indicate highly repetitive outputs). We note that SFT+Tuned produces shorter responses on average (46.4 vs. 77.6 tokens),

which may partially inflate Distinct- n scores. However, the magnitude of the gap suggests the diversity advantage is not solely attributable to response length.

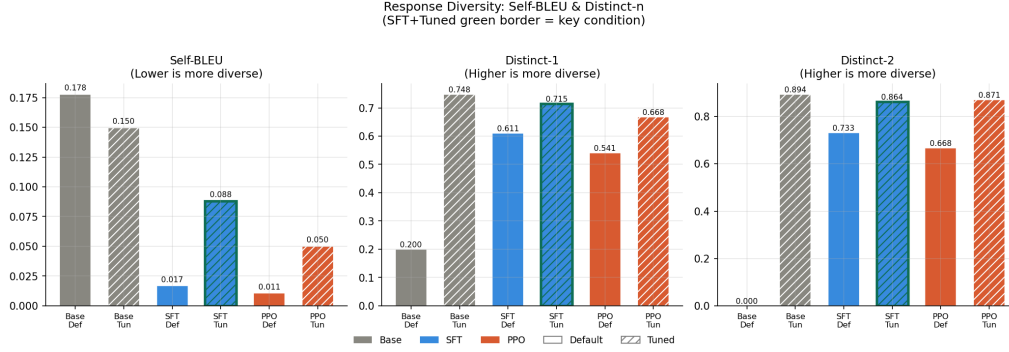


Figure 3: Response diversity across six conditions. Lower Self-BLEU and higher Distinct-1/2 indicate greater diversity. SFT+Tuned (green border) outperforms PPO+Default on all three metrics.

3.3 Hardware Efficiency and Latency

SFT+Tuned averages 0.767s per response, adding only 0.202s over the unoptimized SFT baseline (Figure 4). Peak VRAM remains nearly identical across all conditions (0.629–0.686 GB). PPO+Default requires comparable latency (0.716s) with no efficiency advantage, confirming that Tuned decoding achieves a similar cost profile without the cost of full RLHF training.

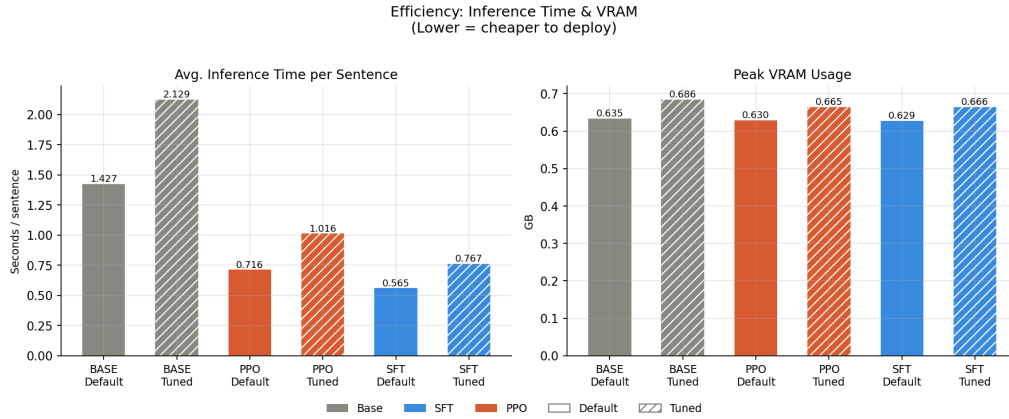


Figure 4: Inference latency and peak VRAM usage comparison. The overhead of Tuned decoding is negligible compared to the computational cost of PPO training.

3.4 Reward Model Reliability

A critical caveat is the strong length bias in our reward model [10]: response length correlates with RM Reward at $r = 0.719$ ($p < 0.001$), suggesting the model rewards longer outputs regardless of content. BLEU [8] and ROUGE scores show no significant correlation with RM Reward ($r \approx 0.30$ – 0.34 , n.s.),

confirming limited discriminability. Diversity and efficiency metrics therefore provide more reliable evidence for evaluating our proposed strategy.

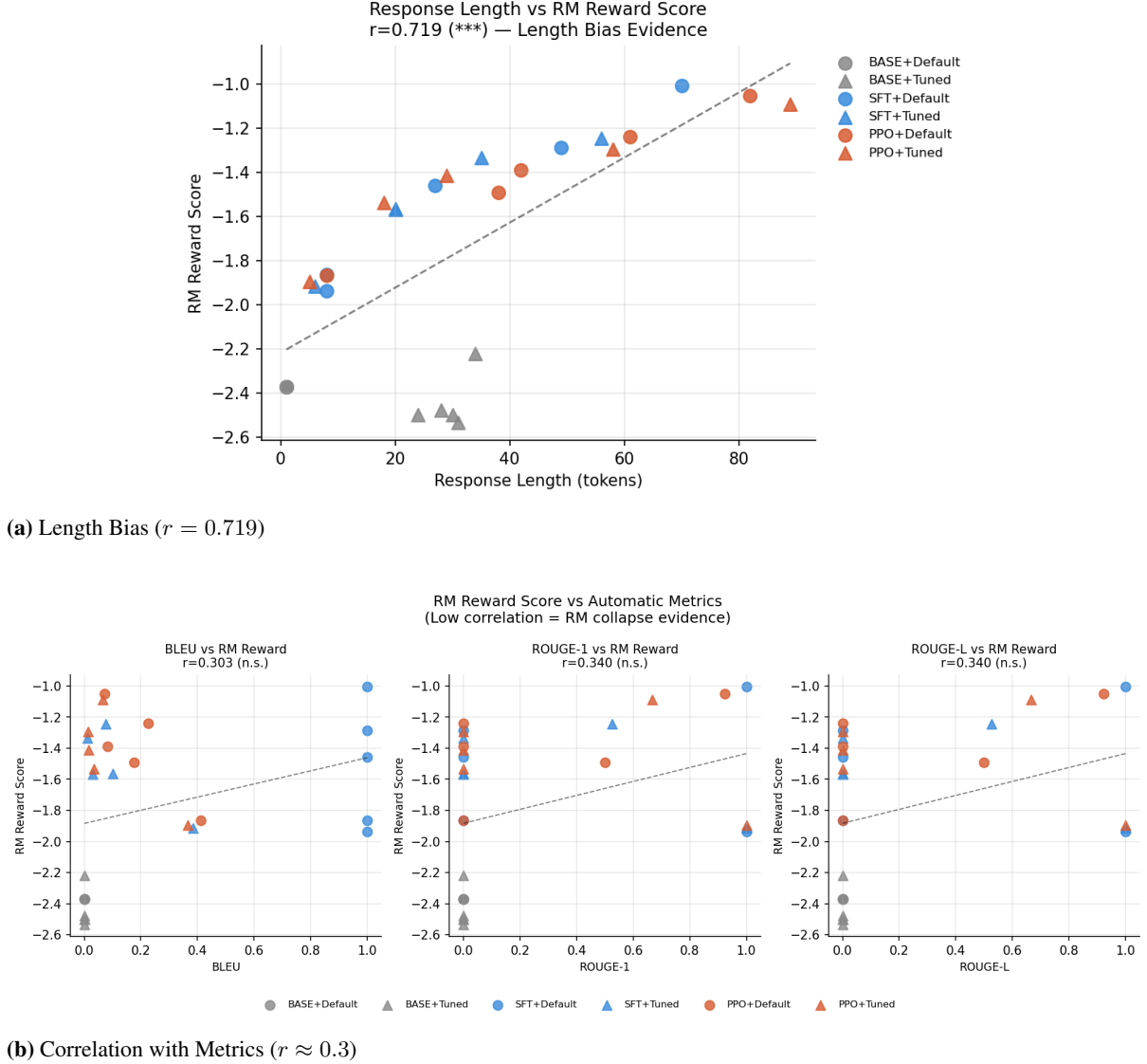


Figure 5: Analysis showing that a strong length bias ($r = 0.719$) contrasts with weak BLEU/ROUGE correlation ($r \approx 0.3$), confirming limited RM discriminability.

4 Discussion and Conclusion

Our study suggests that in resource-constrained environments, inference-time decoding control can serve as a practical alternative to the full RLHF pipeline. While PPO-based alignment achieved marginal gains in proxy reward scores, we identified a strong **length bias** ($r = 0.719$) in the reward model, suggesting that such scores may not reliably reflect response quality in small-scale settings. In contrast, SFT+Tuned approximated PPO-level reward scores while achieving a 29.3% improvement in lexical diversity (Distinct-2) with a latency overhead of only 0.202s. Together, these results indicate that decoding optimization offers a more robust trade-off between alignment quality and computational

efficiency for small-scale Korean LLMs.

Future work could explore more objective evaluation metrics less sensitive to response length, as well as higher-capacity foundation models and better-calibrated reward models, to further validate and extend these findings.

References

- [1] Ouyang, L., Wu, J., Jiang, X., et al. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 27730–27744.
- [2] Airobotlab. (2023). KoChatGPT: Korean ChatGPT training dataset and pipeline. <https://github.com/airobotlab/KoChatGPT>
- [3] SK Telecom. (2019). KoGPT2: Korean GPT-2 model. <https://sktelecom.github.io/project/kogpt2/>
- [4] Holtzman, A., Buys, J., Du, L., Forbes, M., and Choi, Y. (2020). The curious case of neural text degeneration. In *Proceedings of International Conference on Learning Representations (ICLR)*.
- [5] Li, J., Galley, M., Brockett, C., Gao, J., and Dolan, B. (2016). A diversity-promoting objective function for neural conversation models. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 110–119.
- [6] Zhu, Y., Lu, S., Zheng, L., et al. (2018). Texus: A benchmarking platform for text generation models. In *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1097–1100.
- [7] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- [8] Papineni, K., Roukos, S., Ward, T., and Zhu, W. (2002). BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 311–318.
- [9] Lin, C. Y. (2004). ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81.
- [10] Singhal, P., Forrest, T., Long, D., and Novikova, J. (2023). A long way to go: Investigating length correlations in RLHF. *arXiv preprint arXiv:2310.03716*.